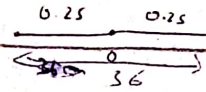


CHAPTER No 15

PROBLEM No 1

GIVEN DATA:-

$$T = 0.25s$$



a) $T = 0.5s$ b) $f = \frac{1}{T} = 2s^{-1}$

c) $x_m = \frac{36}{2} = 18cm$

PROBLEM No 2

GIVEN DATA:-

$$m = 0.12kg$$

b) $k = ?$

$$x_m = 8.5cm = 0.085m$$

a) $F_{max} = ?$

$$T = 0.2s$$

SOLUTION:-

$$\omega = \frac{2\pi}{T} \Rightarrow \frac{2\pi}{0.2} = 31.41$$

$$\omega = \sqrt{k/m} \Rightarrow k = \omega^2 m = 986.96 \times 0.12 = 118.4$$

$$F = -kx_m \Rightarrow -10.06N$$

PROBLEM No 8

$$\phi = ?$$

$$x_m = 6cm$$

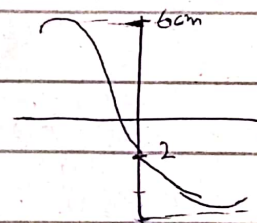
$$x = x_m \cos(\omega t + \phi)$$

when $t = 0$ then $x = -2$

$$-2 = 6 \cos \phi$$

$$\cos \phi = -\frac{1}{3}$$

$$\phi =$$



PROBLEM No 28

$$m = 2 \text{ kg}$$

$$U(x) = bx^2$$

$$U_s = 2 \text{ J}$$

$$v = 0.85 \text{ m/s}$$

$$x = 15 \text{ cm}$$

we know that

$$K.E = \frac{1}{2} mv^2$$

$$= (0.5)(2)(0.85)$$

$$K.E = 0.7225$$

At extreme position

$$x = x_m$$

$$P.E = \max = \max K.E \text{ or } K.E \text{ at mean}$$

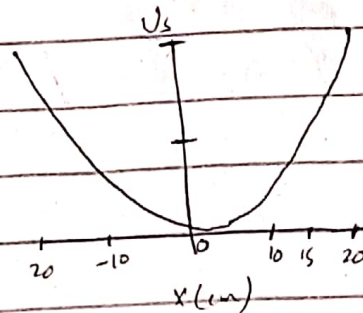
$$P.E_{\max} = 0.7225$$

$\Rightarrow x_m$ has been already reached at

$$U = bx^2$$

$$0.7225 = (5 \times 10^{-3}) x^2$$

$$x = 12.02 \text{ cm}$$



we know that

$$P.E = \frac{1}{2} kx^2$$

$$\text{and given } P.E = U(x) = bx^2$$

let us take $x = 10 \text{ cm}$

so by graph we know

that at $x = 10$ $U = 0.5$

so in eq ①

$$0.5 = b(10)^2$$

$$b = 5 \times 10^{-3}$$

now in $x = 15 \text{ cm}$

$$U(15) = (5 \times 10^{-3})(15)$$

$$U = 1.225$$

$$\omega = \sqrt{\frac{k}{m}}$$

$$\omega =$$

PROBLEM No 32

$$K = ?$$

$$K.E_s = 4J$$

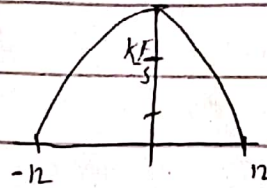
$$K.E_{max} = 6J$$

we know that

$$K.E_{max} = \left(\frac{1}{2}\right) (k x^2)$$

$$6 = (0.5) k (0.12)^2$$

$$k = 833 \text{ N/m}$$



PROBLEM No 36

$$\phi = \pi/6 \text{ rad}$$

$$K.E/P.E = 1$$

$$t = 0$$

$$\frac{K.E}{P.E} = \frac{\frac{1}{2} m v^2}{\frac{1}{2} k x^2}$$

$$= \frac{m \omega^2 x_m^2 \sin^2(\omega t + \phi)}{k x_m^2 \cos^2(\omega t + \phi)}$$

$$= \frac{m \omega^2 \sin^2(\omega t + \phi)}{k \cos^2(\omega t + \phi)}$$

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$$= \frac{m \omega^2 \sin^2(\omega t + \phi)}{k \cos^2(\omega t + \phi)}$$

$$= \frac{m \omega^2 \sin^2(\omega t + \phi)}{k \cos^2(\omega t + \phi)} = \tan^2(\omega t + \phi)$$

$$= \tan^2(\phi)$$

$$= \tan^2(\phi) \quad \therefore t = 0$$

$$\frac{K.E}{P.E} = \frac{1}{3}$$

$$\therefore \omega = \sqrt{\frac{k}{m}}$$

$$\omega^2 = \frac{k}{m}$$

$$\frac{1}{\omega^2} = \frac{m}{k}$$

QUESTION NO 14

$$m = 2 \text{ kg}$$

$$k = 100 \text{ N/m}$$

$$\text{when } t = 10 \text{ s, } x = 0.129 \text{ m and } v = 3.415 \text{ m/s}$$

$$x_m = ? \quad \text{at } t = 0, \quad v = ?$$

we know that

$$\omega = \sqrt{\frac{k}{m}} = \sqrt{\frac{2}{100}} = 7.07$$

we also know that

$$x = x_m \cos(\omega t + \phi) \quad \text{--- (1)}$$

$$v = -\omega x_m \sin(\omega t + \phi) \quad \text{--- (2)}$$

by putting values in (1) and (2)

$$0.129 = x_m \cos(7.07 + \phi) \quad \text{--- (1)}$$

$$3.415 = -7.07 x_m \sin(7.07 + \phi) \quad \text{--- (2)}$$

dividing eq (1) and (2)

$$\frac{3.415}{0.129} = \frac{-7.07 x_m \sin(7.07 + \phi)}{x_m \cos(7.07 + \phi)}$$

$$26.47 = -7.07 \tan(7.07 + \phi)$$

$$\boxed{\phi = -5.7}$$

use again eq (1)

$$x = x_m \cos(\omega t + \phi)$$

$$0.129 = x_m \cos(7.07 \times 0 - 5.7)$$

$$\boxed{x_m = 0.129}$$

$$\text{Now } v = -7.07(0.129) \sin(7.07(0) - 5.7)$$

$$= -0.912 (-0.099)$$

$$\boxed{v = 0.091}$$

CHAPTER No 15

PROBLEM No 57

$$x_m' = x_m - 0.03x_m$$

$$x_m' = 0.97x_m$$

at extreme position

$$E = P.E = \frac{1}{2} k x_m^2$$

$$P.E = \frac{1}{2} k x_m^2 - \frac{1}{2} k (0.97x_m)^2$$

$$= \frac{1}{2} k x_m^2 - \frac{1}{2} k (0.94x_m)^2$$

$$= \frac{1}{2} k (x_m^2 - 0.88x_m^2)$$

$$= \frac{1}{2} k x_m (0.06)$$

So 6% is mechanical energy lost in each cycle.

PROBLEM No 58

$$m = 250$$

$$k = 85 \text{ N/m}$$

$$b = 70 \text{ g/s}$$

$$x_a = ? \text{ at end of } 20^{\text{th}} \text{ cycle.}$$